

Timeout-aware GAE: two boundary decisions

01 Collect the rollout

Store rewards, old critic values, terminal flags and final observations



02 Select the bootstrap value

Terminal: zero; timeout: actual final value; continuing: next value



03 Decide whether the trace continues

Stop across either kind of reset; never borrow the next episode's tail



04 Sweep backward through each world

$TD \text{ residual} + \gamma \times \lambda \times \text{continuing advantage tail}$



05 Prepare PPO learning targets

Actor: advantages; critic: advantages + old values; then optimize

Teaching schematic. Bootstrap at a timeout, but do not propagate across its reset.